

Java Quest – User Manual/Directions

Project Repository

[Java Quest GitHub Repository](#)

Java Quest is an educational web-based game designed to teach Java programming logic and loops through adaptive gameplay, quizzes, XP, badges, and interactive coding challenges.

Files Included

The project primarily uses:

- `index.html`
 - Main game structure and screens
 - `style.css`
 - WSSU-themed styling and layout
 - `script.js`
 - Game logic, adaptive learning system, validation, XP, quizzes, badges, and progression
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Option 1 – View the Project on GitHub

1. Open the repository:
[Open Java Quest Repository](#)
 2. Browse the files:
 - `index.html`
 - `style.css`
 - `script.js`
 3. Click any file to view the code directly in GitHub.
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Option 2 – Download as a ZIP File

Step 1 – Open the Repository

Go to:

[Java Quest Repository](#)

Step 2 – Download ZIP

1. Click the green “Code” button
 2. Select “Download ZIP”
 3. Save the ZIP file to your computer
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Step 3 – Extract the ZIP

- Windows:
 - Right-click → “Extract All”
 - Mac:
 - Double-click the ZIP file
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Step 4 – Open the Folder

Open the extracted **Java-Quest** folder.

Option 3 – Clone Using Git

Open a terminal and run:

```
git clone https://github.com/amariecr/Java-Quest.git
```

Then move into the folder:

```
cd Java-Quest
```

Running the Project

Method 1 – Open Directly in Browser

1. Open the project folder
2. Double-click **index.html**

The game should automatically open in your browser.

Method 2 – Run Using VS Code (Recommended)

Step 1 – Open in VS Code

1. Open VS Code
 2. Select:
 - File → Open Folder
 3. Choose the **Java-Quest** folder
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Step 2 – Install Live Server Extension

Install:

[Live Server VS Code Extension](#)

Step 3 – Start the Project

1. Right-click **index.html**
2. Select:
 - “Open with Live Server”

The game will launch in your browser.

How to Use the Game

Step 1 – Enter Student Information

Step 2 – Complete the Pre-Quiz

The pre-quiz:

- Measures current knowledge
 - Suggests a recommended difficulty and hint mode
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Step 3 – Play Java Quest

Players:

- Write Java loop code
- Receive feedback
- Earn:
 - XP
 - Points
 - Badges

The game checks:

- Loop structure
 - Conditions
 - Updates
 - Output logic
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Step 4 – Master the Difficulty

Easy

- Reach 100 XP

Medium & Hard

- Reach 100 XP
 - AND complete all questions at least once
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Step 5 – Complete the Post-Quiz

The post-quiz:

- Measures learning improvement
 - Includes some preview questions from the next difficulty level
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Step 6 – View Final Summary

The summary includes:

- Pre-quiz vs post-quiz improvement
- XP
- Badge achievements

- Hint usage
 - First-try success rate
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Saving & Resuming Progress

The game supports local save functionality.

To Save

Progress saves automatically while playing.

To Resume

1. Reopen the project
 2. Click:
 - **“Resume Saved Game”**
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Features of Java Quest

- Adaptive difficulty system
 - Output-based loop validation
 - Flexible answer checking
 - XP and badge progression
 - Hint system
 - WSSU Rams themed design
 - Mastery-based progression
 - Analytics and progress tracking
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Troubleshooting

Problem: Game does not open

Solution:

- Make sure `index.html`, `style.css`, and `script.js` remain in the same folder
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Problem: Resume button does not work

Solution:

- Ensure browser local storage is enabled
 - Avoid deleting browser storage data
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Problem: Styling is missing

Solution:

- Verify `style.css` is inside the project folder
- Ensure the HTML stylesheet link is correct

Educational Purpose

Java Quest was designed to:

- Increase student engagement
- Improve Java loop logic understanding
- Encourage problem-solving
- Provide measurable learning improvement through adaptive gameplay

Developer Information

Created by:

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Computer Science Major

Project Theme:

- WSSU Rams inspired design
- Educational game-based learning

Repository:

[Java Quest GitHub Repository](#)